

Statistics Roadmap for Machine Learning

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).

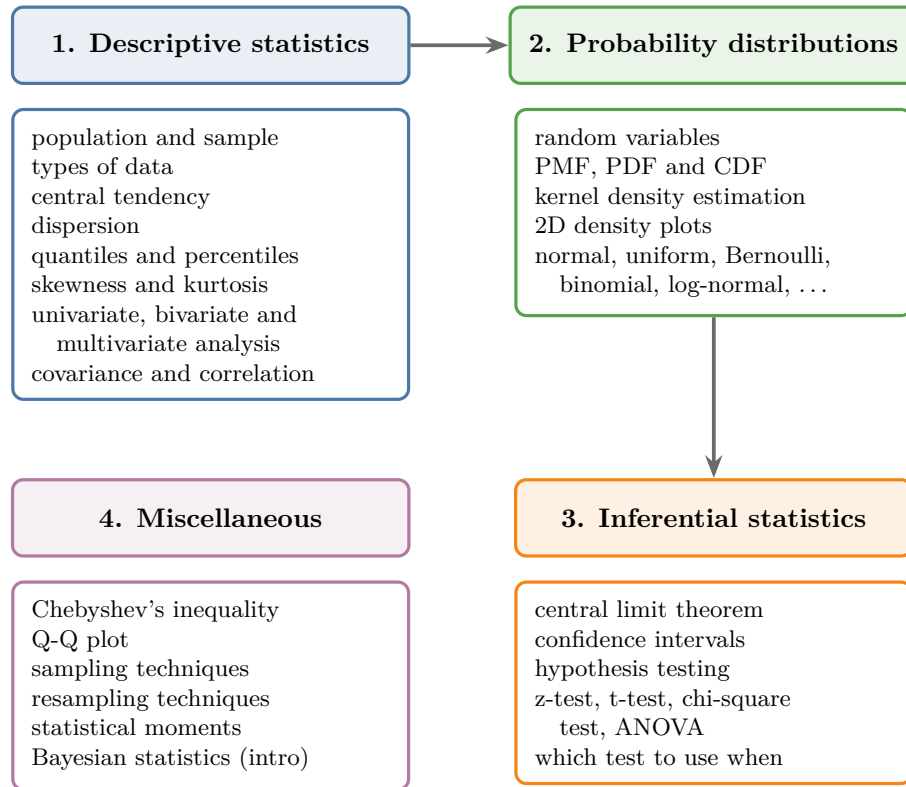


- **Builds on:** Exploratory data analysis ([Note 9](#)); Population, sample, parameter and statistic ([Note 220](#)); Frequency tables ([Note 223](#)); Random variables ([Note 240](#)); Confidence intervals ([Note 280](#)).
- **Leads to:** Measures of central tendency ([Note 221](#)); Variance ([Note 222](#)); Percentiles, quartiles and box plots ([Note 230](#)); Correlation ([Note 231](#)); Normal distribution ([Note 240](#)); Probability mass function (PMF) ([Note 241](#)).

1. Overview

Key point: Statistics (G-1884) for ML splits into four modules: **descriptive statistics** (G-596), **probability distributions** (G-1571), **inferential statistics** (G-944), and a few miscellaneous tools.

Statistics for ML: four modules



each module builds on the one before it;
the miscellaneous topics are studied when an algorithm needs them

Figure 1: The four modules of statistics for ML

Figure 1 shows the whole map. Hypothesis testing, in the inferential module, depends on the probability distributions module. This Note says what each module covers, why ML needs it, and where each topic is taught.

2. Why ML needs statistics

Key point: Statistics is how we summarise data and draw conclusions from it, and many ML algorithms are built on it.

Statistics turns raw data into summaries and decisions. In data work it shows up in three places (Figure 2):

- **Understanding data:** every exploratory analysis is built from statistical summaries and graphs.

- **Algorithms:** linear regression, logistic regression and Naive Bayes all come out of statistics (ISLR ch. 3, §4.3, §4.4.4).
- **Decisions:** checking whether a new medicine, or a new website design, really works better than the old one needs a statistical test (Bruce et al. 2020, ch. 3).

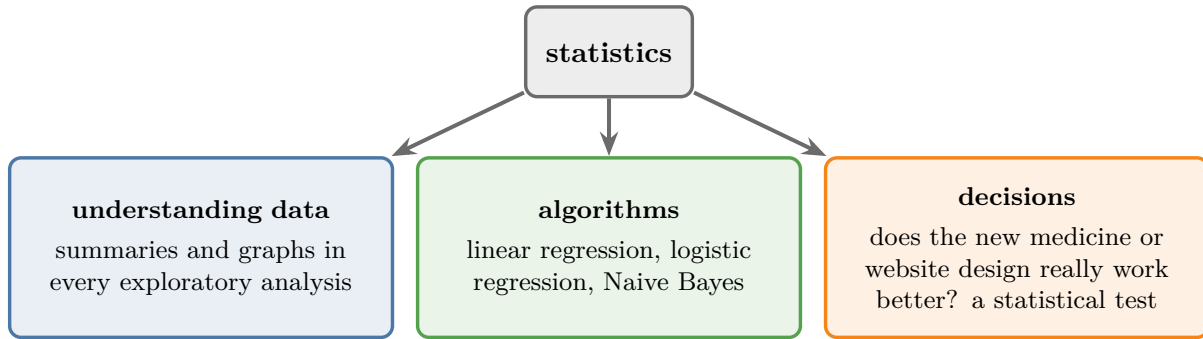


Figure 2: The three places statistics shows up in data work, with an example of each

3. The four modules

Key point: Descriptive statistics summarises the data we have; probability distributions describe the shapes data can take; inferential statistics draws conclusions about data we do not have.

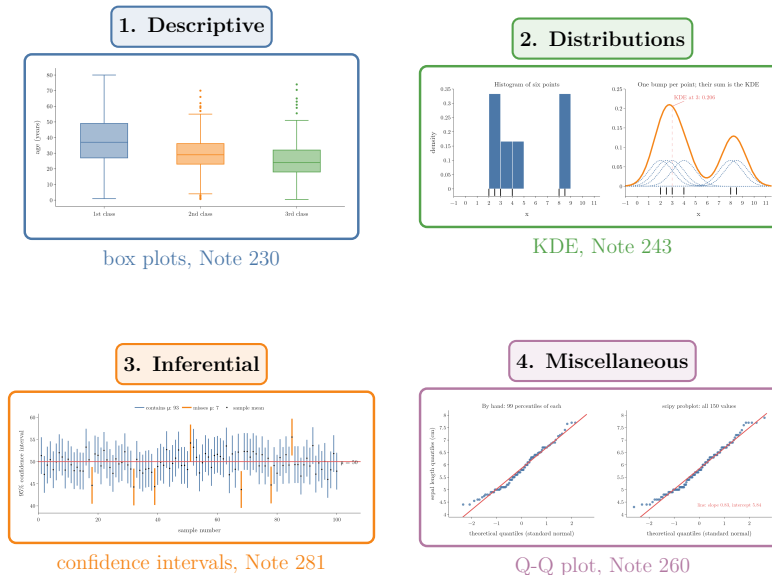


Figure 3: One key figure from each module: box plots of Titanic ages (Note 230), a kernel density estimate built from bumps (Note 243), 100 confidence intervals (Note 281) and a Q-Q plot of iris sepal lengths (Note 260)

Figure 3 previews what each module produces: summaries of the data in hand, a smooth shape for a distribution, a statement about a population with its uncertainty, and a diagnostic plot.

3.1 Descriptive statistics

Key point: Summarise the data we have, with numbers and graphs.

This module describes data that is already in our hands. Its topics, and where each is taught:

Topic	Where it is taught
Population and sample, types of data	What is statistics Note
Central tendency (mean, median, mode, ...)	Measures of central tendency Note
Dispersion (range, variance, standard deviation, ...)	Measures of dispersion Note
Quantiles, percentiles, box plots	Percentiles and box plots Note
Skewness	Univariate analysis Note , skewness Note
Kurtosis	Kurtosis and Q-Q plots Note
Univariate analysis	Univariate analysis Note
Bivariate and multivariate analysis	Bivariate and multivariate analysis Note , frequency tables and graphs Note
Covariance and correlation	Covariance and correlation Note

3.2 Probability distributions

Key point: A probability distribution describes how likely each value of a feature is; the hypothesis tests of the next module depend on it.

This module studies the shapes data can take. Here a **feature** (G-772) is an input variable, one column of the data table, and an **observation** (G-1374) is one record, one row of that table:

- **Random variables** (G-1620), and the functions that describe them: the PMF (for counts), the PDF (for measurements) and the CDF (for running totals of probability).
- **Kernel density estimation**, the smooth curve over a histogram, met in the density plot of the [univariate analysis Note](#).
- **2D density plots**, the same idea for two features at once.
- **Named distributions**: normal, uniform, Bernoulli, binomial, log-normal and others. The normal distribution's 68-95-99.7 rule appears in the [z-score outliers Note](#).

The basic rules of probability are in the [events Note](#) and the [empirical and theoretical probability Note](#); conditional probability is in the [conditional probability Note](#).

3.3 Inferential statistics

Key point: Use a sample to say something about the whole population, and say how sure we are.

This module draws conclusions about a **population** (G-1525) from a **sample** (G-1731):

- **Central limit theorem** (G-364): why averages of samples behave predictably.
- **Confidence intervals:** a range built by a method that captures the true population value in a stated share of repeated samples.
- **Hypothesis testing** (G-913): checking a claim about a population with a sample. The common tests are the z-test, t-test, chi-square test and ANOVA (Bruce et al. 2020, ch. 3), and the module ends with when to use which.

3.4 Miscellaneous topics

Key point: A few tools do not fit one module; we study them when an algorithm needs them.

Topic	Where it is taught
Chebyshev's inequality	a later maths Note
Q-Q plot	Function transformer Note
Sampling techniques	a later maths Note; sampling bias in the challenges in ML Note
Resampling (bootstrap)	Bagging intuition Note
Statistical moments	Kurtosis and Q-Q plots Note
Bayesian statistics	Bayes' theorem Note

4. How to study the roadmap

Key point: About 60 hours, or a month at 2 to 2.5 hours a day; some topics can wait until an algorithm needs them.

The whole roadmap takes roughly 60 hours. At 2 to 2.5 hours a day, that is about one month. Two habits help:

- **Know why each topic matters.** For every topic, note where ML or data analysis uses it.
- **Defer some topics.** A few topics, such as Chebyshev's inequality or Bayesian statistics, can wait until an algorithm needs them. The rest are worth covering now, in the order of the map.

Extra: Good resources for this roadmap:

- **Books:** *An Introduction to Statistical Learning* (James, Witten, Hastie, Tibshirani; statistics through ML), *Practical Statistics for Data Scientists* (Bruce, Bruce, Gedeck; practical, with Python and R), *Think Stats* (Downey; short, Python-

based).

- **Online lessons:** Starmer, J., StatQuest, *Statistics Fundamentals*, statquest.org.
- **Revision:** a one-page statistics cheat sheet, once the topics are learned.

5. Summary

Module	Question it answers	Main topics
Descriptive statistics	What does the data we have look like?	central tendency, dispersion, percentiles, graphs, correlation
Probability distributions	What shapes can data take?	PMF, PDF, CDF, normal and other distributions
Inferential statistics	What can a sample tell us about the population?	central limit theorem, confidence intervals, hypothesis tests
Miscellaneous	Extra tools	Q-Q plot, sampling, resampling, moments, Bayes

- Descriptive statistics comes first; inferential statistics depends on probability distributions.
- About 60 hours covers the whole roadmap.

6. Sources

Built from

- CampusX, “Statistics Roadmap for Data Science and Data Analysis | Complete Guide | Full Resources | CampusX”, YouTube, https://www.youtube.com/watch?v=2GV_ouHBw30

Other references

- ISLR: James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed. Springer. Chapter 3 (linear regression), sections 4.3 (logistic regression) and 4.4.4 (naive Bayes).
- Bruce, P., Bruce, A. and Gedeck, P. (2020). *Practical Statistics for Data Scientists*, 2nd ed. O’Reilly. Chapter 3, Statistical Experiments and Significance Testing.

7. Key terms

This roadmap adds no terms of its own: each term is defined in the Note that teaches it, and listed in the [glossary](#).